Field Guide

Data Curation and Integration: everything from the tutorial, in one place to keep.

Five questions to ask any dataset

Question	Ask	Why it is harder in rural places
Where	What place does this number describe? County, ZIP code, patient's home, or clinic's address?	People often live in one county and get care in another.
When	What year does the data describe? That can differ from the year it was released.	Small places often get only five-year estimates.
What	What exactly is being counted? Is it measured, from a survey, or modeled?	Words like visit, uninsured, and access change between sources.
How sure	What is the margin of error? Are any values hidden?	Fewer people means wider ranges and more hidden values.
Who's missing	Who is not in this data? Does a blank mean zero or unknown?	Fewer monitors, fewer survey responses, and no record of people who never reached care.

Three kinds of numbers

A **measured** number comes from a monitor or an actual count. A **survey** number comes from a sample of people who answered questions. A **modeled** number is calculated from other data. All three are useful. Label which one you have.

Your first join, in four steps

1. Put both tables in one workbook. Format the county code column as text before you paste, so leading zeros stay. 2. In the first table, add a new column and enter the formula below. 3. Copy it down, then search for NOT FOUND. 4. Add columns for source, year, and type.

```
=XLOOKUP(A2, Sheet2!A:A, Sheet2!C:C, "NOT FOUND")
```

This takes the code in A2, finds it in column A of the second sheet, and brings back the value from column C. Then check that you have the same number of rows as before.

Using AI: three checks and one rule

Count rows before and after. **Spot-check** three places by hand on the original website. **Ask why**: make it quote the source for any definition, and if it can't, don't use it. **Never** paste information about real patients into a public AI tool, even with names removed.

Tools: what they are and how to get to them

All free. Links checked November 5, 2026. Websites change, so download what you use and write down the date.

Look it up (no download)

Tool	What it is	Why you might need it	How to access it	Watch out
County Health Rankings & Roadmaps	One page for each US county, covering health, income, housing, and environment.	To get to know a place quickly.	countyhealthrankings.org/health-data. Search a county. State data files are on each state page.	Measures mix many years. Open a measure to see its years.
Data Commons	A search engine for public statistics from many agencies, from Google.	When you don't know where a number lives.	datacommons.org. Type a place and a topic.	Check the source listed under each chart.
Rural Data Explorer	Maps of poverty, health workforce, and other measures, from the Rural Health Information Hub.	To compare metro and non-metro counties.	ruralhealthinfo.org/data-explorer	Metro and non-metro are county labels, so a metro county can still be mostly rural.

Download a table someone else combined

Tool	What it is	Why you might need it	How to access it	Watch out
Environmental Justice Index	Ranks every census tract on 36 environmental, social, and health factors. From the CDC.	To get health and environment in one file.	atsdr.cdc.gov/place-health/php/eji/index.html. Use the explorer map or the data download page.	It gives a relative rank, not a measured exposure. A court order has affected this site before, so save a dated copy.
Social Vulnerability Index	Ranks counties and census tracts on 16 census measures such as poverty and access to a car. From the CDC.	To see which communities may need more support.	atsdr.cdc.gov/place-health/php/svi/index.html	Margins of error are included. Use them, especially in small counties.
CDC PLACES	Local health estimates for counties, places, census tracts, and ZIP code areas.	When you need a health number and no local survey exists.	cdc.gov/places. Map at places.cdc.gov; data files at data.cdc.gov.	These are modeled estimates. Pick plain or age-adjusted and stay with it.
Environmental Public Health Tracking Network	Air, water, heat, and other environmental data alongside health data. From the CDC.	To look at environment and health together.	ephtracking.cdc.gov. Open the Data Explorer.	Each measure has its own years and geography. The site can be down for maintenance.
Community-Level Health Database	Yearly files on social, economic, and health conditions at county, ZIP code, tract, and block group level. From the Agency for Healthcare Research and Quality.	When you want many variables in one linkable file.	ahrq.gov/data/innovations/clh-data.html	Files are large, so start with a county file. Read the codebook.

Go to the original source

Tool	What it is	Why you might need it	How to access it	Watch out
CDC WONDER	Death records and other public health statistics.	For deaths by cause, place, and year.	wonder.cdc.gov	Local counts under 10 are hidden. Rates from small counts are flagged or hidden as unreliable, and the rule varies by file.
data.census.gov	Census counts and American Community Survey estimates.	For population, income, housing, and insurance.	data.census.gov. Search a county name.	Small counties get five-year estimates. Keep the margin of error.
USDA rural codes	Rural-Urban Continuum Codes for counties and Rural-Urban Commuting Area Codes for census tracts and ZIP code areas.	To show which definition of rural you are using.	ers.usda.gov/data-products/rural-urban-continuum-codes	Codes change with each census, so note the version.
EPA AirData	Air quality data from outdoor monitors.	To check whether a place has a real monitor.	epa.gov/outdoor-air-quality-data	Most counties have no monitor of their own, so most county figures are estimates.

Combine it yourself

Tool	What it is	Why you might need it	How to access it	Watch out
XLOOKUP in Excel or Google Sheets	A formula that pulls matching values from one table into another.	For your first join. No coding needed.	Help pages: support.microsoft.com (search XLOOKUP) and support.google.com/docs/answer/12405947	Needs Excel 2021 or later, Microsoft 365, or Google Sheets. Format county codes as text first, so leading zeros stay.
OpenRefine	A free desktop tool for cleaning messy data.	To fix inconsistent spellings and categories.	openrefine.org	Keep the original file untouched.
Datawrapper	A free website for maps and charts.	To make a county map without coding.	datawrapper.de	Don't upload private data. Say which year the map shows.
QGIS	A free, full mapping program.	For more advanced map work, such as drive times.	qgis.org	It takes longer to learn. Try Datawrapper first.
HUD-USPS ZIP Code Crosswalk	Files that convert ZIP codes to counties and census tracts.	When one table uses ZIP codes and another uses counties.	huduser.gov/portal/datasets/usps_crosswalk.html. Free login required.	A ZIP code can span several counties, so the conversion is approximate.

Prompts you can copy

Read the fine print
You are helping a beginner. Using only the attached document, tell me what place this data describes, what years it covers, how the main number is calculated, and who is left out. Quote the section for each answer. If the document doesn't say, answer “not stated.”

Match columns
Here are the column names and definitions from two public datasets. Tell me which columns can be matched, which only look similar, and what I should check before joining them. Don't guess at a definition I haven't given you.

Write the formula
Write a Google Sheets formula that joins Table B onto Table A using a five-digit county FIPS code stored as text, keeping leading zeros. Explain each part in plain English, and give me two ways to check the result.

Source Card

Make one card for each dataset. Copy this page as often as you need.

Dataset, publisher, and link	
Date I downloaded it	
Where	
When	
What	
How sure	
Who's missing	

Example. CDC PLACES, 2025 release, diabetes among adults. **Where:** county, Greene, Alabama (FIPS 01063). **When:** 2023 data, released in 2025. **What:** share of adults with diagnosed diabetes, age-adjusted, modeled from a phone survey. **How sure:** 21.1%, likely range 18.5% to 23.7%. **Who's missing:** phone surveys can miss people without a reliable phone.

Plain-language glossary

Age-adjusted. A rate recalculated so that places with older or younger populations can be compared fairly.

Census tract. A small statistical area inside a county, usually 1,200 to 8,000 people.

Curation. Keeping data organized, labeled, and honest about where it came from.

FIPS code. A standard ID number for a place. A county's has five digits: two for the state and three for the county.

Fine particle pollution (PM2.5). Tiny particles in the air that can reach deep into the lungs.

Integration. Combining data from different sources so it describes the same place and time.

Join (or merge). Matching rows in two tables using a column they share.

Margin of error. How far an estimate could reasonably be from the true value.

Metro / non-metro. A county-level label from a federal list, based on city size and commuting ties.

Modeled estimate. A number calculated from other data, not counted or measured in that place.

Suppressed. Hidden on purpose, usually because the count is small. Never read it as zero.

ZIP code area. A census version of a ZIP code, used for statistics. Also called a ZCTA.

Want to go deeper?

Hart, Larson & Lishner (2005). Rural definitions for health policy and research. American Journal of Public Health. Why your definition of rural can change your conclusions.
https://pmc.ncbi.nlm.nih.gov/articles/PMC1449333/

Tiwari, Beyer & Rushton (2014). The impact of data suppression on local mortality rates: the case of CDC WONDER. American Journal of Public Health. How hidden small counts affect rural numbers, and a way to adjust.
https://pmc.ncbi.nlm.nih.gov/articles/PMC4103252/

McKenna et al. (2017). Human intestinal parasite burden and poor sanitation in rural Alabama. American Journal of Tropical Medicine and Hygiene. The first hookworm study from Part 1.
https://pubmed.ncbi.nlm.nih.gov/29016326/

Poole et al. (2023). Cross-sectional study of soil-transmitted helminthiases in Black Belt region of Alabama, USA. Emerging Infectious Diseases. The larger study from Part 1, which found no hookworm in stool samples.
https://pmc.ncbi.nlm.nih.gov/articles/PMC10683802/

Thomas, Dobis & McGranahan (2024). The nature of the rural-urban mortality gap. USDA Economic Research Service. How the rural death-rate gap grew over 20 years.
https://www.ers.usda.gov/publications/108701